Abuzar Khan

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Summary

Data Scientist and AI Engineer with hands-on experience building, evaluating, and deploying supervised machine learning models (regression, classification) using Python and scikit-learn. Proficient in the full modeling lifecycle — from exploratory data analysis and feature engineering through hyperparameter tuning, cross-validation, model evaluation (AUC-ROC, F1, RMSE), and production deployment. Experienced with SQL for data extraction, Git for version control, and Docker for containerized deployment. Passionate about translating complex analytical findings into actionable business insights and building scalable, production-ready AI solutions.

Education

MIT, Aurangabad **2021 – 2025**
Bachelor of Technology (B.Tech) — Artificial Intelligence & Data Science

Maulana Azad College of Arts and Science
Higher Secondary Education (HSC)

Experience

Freelance AI/ML Engineer – Remote **March 2026 – Present**
NCCO.sa, Saudi Arabia

- Architecting an end-to-end internal web application featuring an LLM-based AI chatbot with PDF Q&A, web search, and document assistance; translating business requirements into production-ready ML solutions for a regulated organizational environment.
- Designed and optimized RAG pipelines with prompt engineering strategies, reducing hallucination rate and improving response relevance for domain-specific knowledge base queries.
- Collaborated with business stakeholders to define success metrics, scope project requirements, and communicate analytical findings through structured documentation and demos.

Software Developer Intern – On-site **Apr – Oct 2025**
NRB Bearings Limited

- Developed a GUI-based Tool Crib Management System, writing SQL queries for data extraction and transformation against an SQL Server database to support inventory analytics.
- Built a web-based Kaizen Entry Dashboard with data visualization capabilities, enabling business stakeholders to monitor operational KPIs and make data-driven decisions.
- Proposed and documented automation improvements to digital workflows based on data analysis, presenting recommendations to technical and non-technical audiences.

Skills

Machine Learning: Regression, Classification, Clustering, Random Forest, XGBoost, LightGBM, Gradient Boosting, SVM, Feature Engineering, Hyperparameter Tuning, Cross-Validation, SMOTE, Imbalanced Data Handling
Model Evaluation & Explainability: AUC-ROC, F1, Precision-Recall, RMSE, MAE, R^2, Lift Curves, SHAP Values, Partial Dependence Plots
Deep Learning & NLP: CNN, RNN, TensorFlow, PyTorch, BERT, Word2Vec, NLTK, spaCy, Text Classification, Hugging Face Transformers
Gen AI: Groq LLM (Llama), Gemini 1.5, OpenAI GPT APIs, RAG Pipelines, Prompt Engineering, n8n
Data Analysis & Visualization: EDA, Pandas, NumPy, Matplotlib, Seaborn, Power BI, SciPy
Programming & Frameworks: Python (Scikit-Learn, FastAPI, Flask), SQL (SQL Server), NoSQL (MongoDB), React
MLOps & Deployment: Docker, Git/GitHub, Render (cloud deployment), CI/CD pipelines, Flask, Gunicorn, Experiment Tracking
Cloud & Tools: Render (cloud hosting), Microsoft IIS, Power BI, Excel/DAX

Projects

Credit Card Fraud Detection System | *Python, Flask, scikit-learn, Random Forest, XGBoost* **Link**

- Built and deployed a supervised classification model on a highly imbalanced dataset (284K+ transactions, ~0.17% fraud rate); applied SMOTE and class-weight tuning to handle class imbalance effectively.
- Evaluated models using AUC-ROC (0.97), F1-score (0.91), and precision-recall curves; applied cross-validation and hyperparameter tuning (GridSearchCV) for robust, generalizable model selection.
- Developed Flask-based web application with REST API for real-time fraud predictions; deployed to Render cloud platform with CI/CD integration, ensuring production-ready reliability.
- Implemented SHAP values for model explainability, enabling interpretable fraud predictions for both technical and non-technical stakeholders.

E-Commerce Business Intelligence Dashboard | *Power BI, SQL, DAX, Excel* **Link**

- Developed an interactive Power BI dashboard for e-commerce analytics by writing SQL queries to extract and transform structured sales data from relational databases.
- Delivered insights on sales trends, profit margins, customer segmentation, and payment behavior; utilized by business stakeholders for data-driven decision-making and sales forecasting.

Real-Time Facial Expression Detection Model | *Python, OpenCV, TensorFlow/Keras*

- Trained a CNN-based classification model on the FER-2013 dataset (35K+ images, 7 expression classes); applied data augmentation, dropout regularization, and hyperparameter tuning achieving 92% validation accuracy.
- Evaluated model performance using F1-score and confusion matrix per class; implemented real-time inference pipeline on live webcam streams for HCI and emotion-aware AI applications.

Certifications

Career Essentials in Data Analysis by Microsoft and LinkedIn **Link**

Data Analytics Job Simulation by Deloitte Australia (via Forage) **Link**

Internship Common Aptitude Test by LABMATIC **Link**

Participated in Mumbai Hack 2025 — Largest Agentic AI Hackathon **Link**